

ViTPrompt: Training-Free Prompt Refinement with Visual Tokens for Open-Vocabulary Detection

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Abstract

*Test-Time Adaptive Object Detection (TTAOD) aims to maintain detection performance under distribution shifts without retraining. While recent vision-language models enable open-vocabulary detection, existing TTAOD methods—whether closed-set or open-vocabulary—focus solely on improving classification confidence and ignore the degradation of bounding box localization. To address this critical gap, we propose **ViTPrompt** (Visual Token-Prompting), a training-free framework that jointly refines both bounding boxes and class scores at test time. Our key insight is to augment the original text prompt with instance-aware visual tokens extracted from high-confidence detections in an initial forward pass. This enhanced prompt is then fed into a second inference stage, where the cross-modal decoder produces updated box coordinates and more accurate classification logits through enriched semantic context. ViTPrompt requires no back-propagation, parameter updates, or external memory, making it highly efficient for real-time deployment. Experiments on multiple out-of-distribution benchmarks show that ViTPrompt achieves state-of-the-art performance, with consistent gains in both localization accuracy and classification fidelity, demonstrating its effectiveness as a holistic solution for open-vocabulary TTAOD. We will release our code at <https://github.com/buerzlh/Test-time-Adaptation-for-Object-Detection>.*

1. Introduction

Object detection has become a foundational capability in real-world vision systems, powering applications from autonomous navigation to industrial inspection [24, 25, 40]. However, models trained on curated datasets often degrade significantly when deployed under distribution shifts—such

as changes in weather, lighting, or scene context—due to the mismatch between training and test domains [18]. Test-Time Adaptive Object Detection (TTAOD) addresses this by adapting detectors *during inference* without access to source data or labels [20].

Most existing TTAOD approaches build upon closed-set detectors like Faster R-CNN [32], limiting their applicability to predefined categories and failing to generalize to open-vocabulary settings [22]. More critically, these methods [4, 13, 19, 36, 41, 44] focus almost exclusively on improving *classification confidence*—through feature alignment, entropy minimization, or classifier recalibration—while leaving *bounding box predictions untouched*. As a result, even when classification scores increase, the detector may produce highly confident yet spatially inaccurate localizations, especially under severe domain shifts (e.g., foggy scenes causing scale distortion or occlusion-induced boundary errors).

The rise of vision-language models (VLMs) offers a path toward open-vocabulary TTAOD. Models like Grounding DINO [22] enable detection via arbitrary text prompts by aligning image regions with language semantics through cross-modal attention. Yet, existing TTAOD methods [1, 55] based on VLM treat prompts as static inputs, ignoring the rich visual evidence available at test time. When the initial prompt poorly matches the target instance—due to ambiguity, brevity, or domain gap—the resulting alignment is suboptimal, leading to both misclassification and imprecise localization.

To overcome these limitations, we propose **ViTPrompt** (Visual Token-Prompting), a training-free, two-stage inference framework that jointly refines bounding boxes and classification scores by augmenting textual prompts with instance-aware visual tokens. Our key insight is simple yet effective: high-confidence detections from an initial forward pass contain valuable visual cues that can be projected into the language space to enrich the semantic con-

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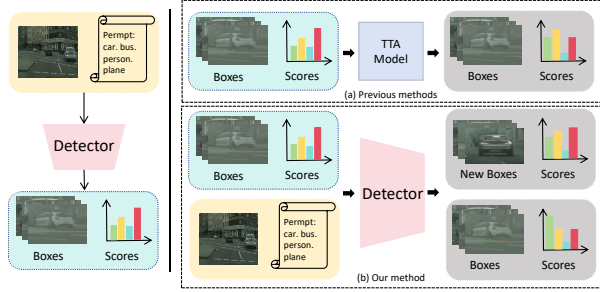


Figure 1. **How TTA methods operate after the initial detection pass.** Given the output of a base open-vocabulary detector (e.g., Grounding DINO)—a set of bounding boxes and confidence scores—test-time adaptation (TTA) methods aim to improve robustness without retraining. (a) *Conventional TTA* treats this output as fixed: it only re-scores or refines the *original proposals* using the same static text prompt, and cannot propose new spatial locations. (b) *Our ViTPrompt* goes beyond post-hoc adjustment. It uses RoI-aligned visual tokens extracted from the initial detections as input to a *second-stage inference*, which produces **new candidate boxes and scores**. This enables true instance-level adaptation, correcting misaligned or missing objects from the first pass.

text for a second inference. This enhanced prompt enables the detector’s cross-modal decoder to re-attend to more relevant image regions, yielding not only better-aligned token similarities but also updated box coordinates through the same decoding mechanism. Crucially, ViTPrompt requires no backpropagation, parameter updates, or external memory—making it efficient and practical for real-time deployment. As illustrated in Figure 1, while prior TTAOD methods only boost classification confidence with fixed boxes, ViTPrompt holistically improves both localization accuracy and semantic fidelity.

Our contributions are threefold:

- (1) We identify and address a critical gap in TTAOD: the lack of bounding box adaptation in existing methods;
- (2) We introduce ViTPrompt, the first training-free framework that jointly refines boxes and scores via prompt augmentation with self-generated visual tokens;
- (3) We achieve state-of-the-art performance on multiple out-of-distribution benchmarks, demonstrating consistent gains in both localization and classification.

2. Related Work

2.1. Open-Vocabulary Object Detection

Traditional object detectors are restricted to a fixed set of categories defined during training [2, 31], which severely limits their applicability in open-world scenarios. The emergence of vision-language models (VLMs) [30, 53] has enabled *open-vocabulary object detection*—the ability to detect arbitrary object categories specified via natural language at test time.

Early VLMs focused more on classification problems. CLIP [30] pioneered large-scale contrastive pretraining on image-text pairs, demonstrating remarkable zero-shot generalization. This inspired a series of VLMs such as ALIGN [15], FILIP [45], PaLI [3], LiT [51], and ZeroVL [5], which further improved cross-modal alignment through scaling, fine-grained interaction, or efficient training strategies. Recently, more and more VLMs are transferring CLIP’s knowledge to conventional detectors via distillation (e.g., ViLD [10]) or pseudo-labeling (e.g., DetCLIP [46]). More recent approaches integrate VLMs directly into detection pipelines. OV-DETR [50] uses CLIP embeddings as queries in DETR [2]; GLIP [17] unifies detection and phrase grounding through multi-task learning; Grounding DINO [22] achieves state-of-the-art performance by combining language-guided query selection with multi-scale feature fusion.

Our work builds exclusively on Grounding DINO as the base detector. Unlike earlier frameworks that rely on external VLMs like CLIP for classification, Grounding DINO natively supports region-to-text alignment within its architecture, making it an ideal foundation for test-time adaptation in open-vocabulary settings.

2.2. Test-Time Adaptation for Robust Perception

Test-time adaptation (TTA) aims to maintain model performance under distribution shift by adapting to unlabeled test samples online, without accessing original training data or permanently updating model parameters.

In image classification, TTA methods include entropy minimization [27, 42, 54], batch normalization recalibration [28, 38], pseudo-labeling [13, 14], and consistency regularization [29, 39, 43]. With VLMs, prompt-tuning-based TTA emerged: TPT [35] optimizes text prompts via test-time backpropagation; DiffTPT [9], C-TPT [48], and PromptAlign [34] enhance robustness through augmentation or distribution matching. However, these methods require gradient computation, hindering real-time deployment. To address this, recent works propose *training-free* TTA: DOTA [11] estimates test distribution online; MTA [49] manages augmented views without gradients; Zero [8] uses hard pseudo-labels via temperature scaling; TDA [16] refines predictions through a lightweight cache.

Despite these advances, TTA for *object detection* remains nascent—particularly for open-vocabulary models. Existing TTAOD methods (e.g., STFAR [4], MemCLR [37]) target closed-set detectors like Faster R-CNN and rely on computationally expensive backpropagation.

Recently, VLOD-TTA [1] leverages IoU-weighted entropy minimization on spatially coherent proposals and image-conditioned selection of informative prompts to improve robustness under domain shifts. While BCA [56] and its enhanced variant BCA+ [55] introduce a training-

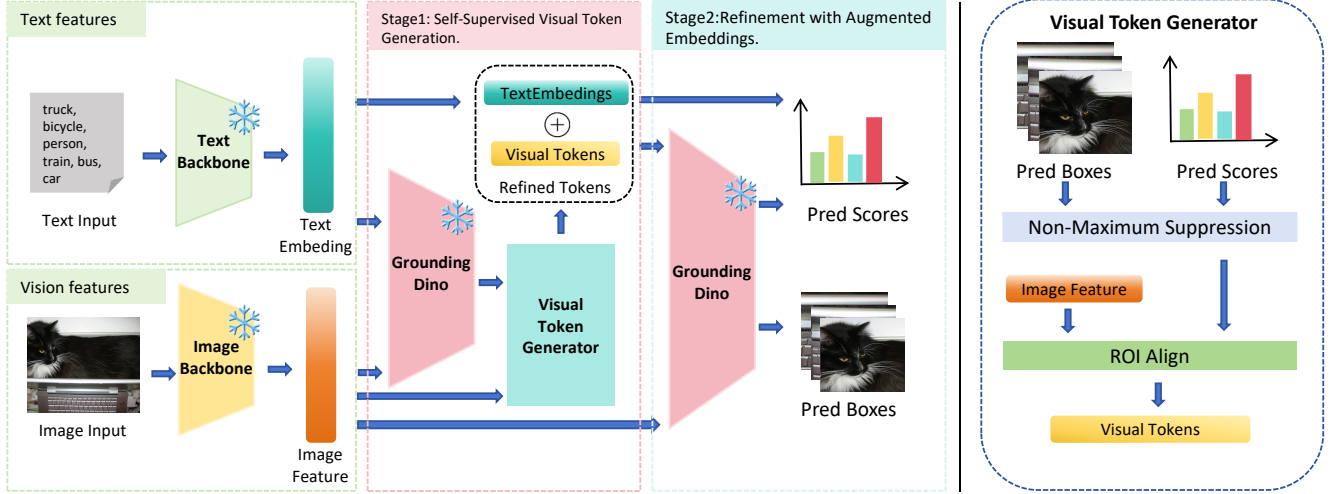


Figure 2. **Overview of ViTPrompt.** Our method operates in two stages to generate instance-adaptive visual prompts for open-vocabulary detection. **Stage 1 (Self-Supervised Visual Token Generation):** Given an image and a text prompt (e.g., “truck . bicycle . person ...”), the Grounding DINO model generates initial object proposals (boxes) and aligns visual tokens to these regions via RoI-Align. **Stage 2 (Refinement with Augmented Embeddings):** The RoI-aligned visual tokens are fused with the original text embeddings to create refined, instance-specific prompts. These dynamic prompts are then used to re-score the initial predictions, yielding more robust detection results under domain shift.

free framework that employs a Bayesian cache to jointly estimate likelihood (via region proposal embeddings) and prior (via contextual category beliefs), achieving strong robustness without gradient updates. While effective, BCA+ operates at the category level and does not refine language queries based on instance-specific visual evidence, limiting its adaptability to ambiguous or corrupted proposals.

To overcome this limitation, we propose ViTPrompt, a novel training-free TTA method that enables *instance-level* visual prompting. By dynamically aligning RoI-extracted visual tokens with language-conditioned queries, ViTPrompt achieves fine-grained, context-aware adaptation for each detection proposal—without modifying model weights or computing gradients.

3. Method

3.1. Preliminaries

Problem setting. In this work, we tackle the problem of test-time adaptation in open-vocabulary object detection, building upon a pretrained vision-language foundation model, Grounding DINO [22]. For a stream of incoming test images $\{\mathbf{x}_i\}_{i=1}^n$ presented in sequence, the detector output, for each $\mathbf{x}_i$, a collection of bounding boxes $\{\mathbf{b}_{ij}\}_j$ and corresponding category labels $\{c_{ij}\}_j$ drawn from a set of K specified classes, all in an online fashion without delay. Our objective is to improve the detection performance, as quantified by mean Average Precision (mAP), across the test set $\{\mathbf{x}_i\}_{i=1}^n$ in the presence of domain shifts, while ensuring computational efficiency suitable for real-time scenarios.

Grounding DINO. Grounding DINO [22], a leading vision-language model, excels in open-vocabulary object detection by aligning visual and textual representations, enabling flexible detection across diverse categories. It serves as a strong foundation for TTAOD tasks, where we use it to produce initial outputs including bounding boxes $\{\mathbf{b}_{ij}\}_j$, class predictions $\{\mathbf{p}_{ij}^{init}\}_j$, and feature embeddings $\{\mathbf{f}_{ij}\}_j$.

For an input image $\mathbf{x}_i \in \mathbb{R}^{3 \times H \times W}$ and text prompts $\mathbf{T} = \{[t_k]\}_{k=1}^K$ (e.g., “cat . dog .”), Grounding DINO employs a dual-encoder setup. The visual encoder ϕ_v , based on Swin Transformer [23], extracts multi-scale visual features $\mathbf{F}_v' = \{\mathbf{F}_l'\}_{l=1}^L$, where L denotes the number of scales, and $\mathbf{F}_1 \in \mathbb{R}^{d \times H' \times W'}$ represents the largest-scale feature map (with H', W' as downsampled dimensions and d as channels). The text encoder ϕ_t , derived from BERT [6], generates textual embeddings $\mathbf{F}_t' = \phi_t(\mathbf{T})$. These are fused via attention-based enhancers to yield enhanced visual features $\mathbf{F}_v = \psi_v(\mathbf{F}_v', \mathbf{F}_t') \in \mathbb{R}^{N_I \times d}$ and textual features $\mathbf{F}_t = \psi_t(\mathbf{F}_v', \mathbf{F}_t') \in \mathbb{R}^{K \times d}$, with $d = 256$ and N_I as image tokens. $\mathbf{F}_t$ functions as a dynamic classifier. A language-guided query selection then picks $N = 900$ queries $\{\mathbf{q}_j\}_{j=1}^N = \mathbf{F}_v[\mathbf{I}]$, with indices determined by $\mathbf{I} = \text{Top}_N(\text{Max}^{(-1)}(\mathbf{F}_v \mathbf{F}_t^\top))$, where $\text{Max}^{(-1)}$ computes the maximum along the last dimension and Top_N extracts the top N query indices.

A DETR-style decoder [52] refines these queries via cross-modality interactions with $\mathbf{F}_v$ and $\mathbf{F}_t$, yielding refined proposal embeddings $\{\mathbf{f}_q\}_{q=1}^N \in \mathbb{R}^d$. From these, class logits $\mathbf{l}_q \in \mathbb{R}^K$ are computed as $\mathbf{l}_q = \sigma(\mathbf{f}_q \mathbf{F}_t^\top)$, where σ

denotes the sigmoid function, score values $s_q = \max(l_q)$ are used for sorting in non-maximum suppression (NMS), and bounding boxes $\{b_q\}_{q=1}^N \in \mathbb{R}^4$ (as $[x, y, w, h]$) are regressed using a dedicated head. High-confidence predictions are then filtered where $s_q \geq \tau_1$, producing the final set $\{(b_q, p_q^{init}, s_q)\}_{q=1}^Q$, with p_q^{init} as softmax-normalized logits l_q .

3.2. Visual Token-Augmented Prompting

Despite its impressive open-vocabulary detection performance, Grounding DINO relies on static text prompts during inference, which can exacerbate semantic misalignment under domain shifts—such as adverse weather conditions or shifts from synthetic to real-world data. This misalignment arises primarily from the fixed textual embeddings F_t , which fail to adapt to altered visual contexts, leading to sub-optimal query selection: the initial queries are chosen based on maximum similarity between visual features and these rigid embeddings, often resulting in low-confidence classifications and imprecise bounding boxes. Moreover, conventional TTAOD approaches exacerbate this issue by merely recalibrating prediction scores without refining the underlying localization or query selection processes.

To address these shortcomings, we introduce **ViTPrompt**, a training-free, two-stage inference pipeline that dynamically augments the textual embeddings F_t with instance-specific visual tokens cropped from the largest-scale multi-scale feature map $F_1 \in \mathbb{R}^{d \times H' \times W'}$ in the detector’s initial output. By enriching F_t in this manner, we enable a refined query selection that better bridges the domain gap, achieving comprehensive improvements in both classification confidence and bounding box accuracy through just one extra forward pass.

Stage 1: Self-Supervised Visual Token Generation.

Our ViTPrompt framework commences with a standard forward pass through Grounding DINO on the test image x_i , producing initial detections as detailed previously: a set of proposals $\{(b_q, p_q^{init}, s_q)\}_{q=1}^Q$. To curate reliable instances for augmentation, we select the top- K queries by sorting in descending order of s_q :

$$\mathcal{S} = \text{argsort}_{desc}(\{s_q\}_{q=1}^Q), \quad (1)$$

$$\{b_k, p_k^{init}, s_k\}_{k=1}^K = \{(b_{\mathcal{S}[k]}, p_{\mathcal{S}[k]}^{init}, s_{\mathcal{S}[k]})\}_{k=1}^K,$$

where $\text{argsort}_{desc}(\cdot)$ denotes descending argument sort, and K (e.g., 10) is a hyperparameter balancing coverage and quality.

Next, we refine the entire set of proposals by focusing on classes observed in the top- K : let $\mathcal{C} = \{\arg \max(p_k^{init}) \mid k = 1, \dots, K\}$ be the set of salient classes. We discard all proposals where $\arg \max(p_q^{init}) \notin \mathcal{C}$, restricting adaptation to domain-relevant semantics. For each class $c \in \mathcal{C}$, we then apply class-specific Non-Maximum Suppression

(NMS) to the remaining proposals of class c , using the top- K boxes of that class as anchors. For a candidate box b_q (with $\arg \max(p_q^{init}) = c$) and an anchor b_k (also class c), we compute IoU:

$$IoU(b_q, b_k) = \frac{(b_q \cap b_k)}{(b_q \cup b_k)}, \quad (2)$$

and suppress b_q if $IoU > \theta$ (e.g., $\theta = 0.6$) and $s_q < s_k$. This process yields a refined set $\{(b_m, p_m^{init}, s_m)\}_{m=1}^M$ (with $M \leq Q$), prioritizing high-quality, non-redundant instances aligned with top predictions.

For this post-refinement set, we extract visual tokens from the largest-scale feature map $F_1 \in \mathbb{R}^{d \times H' \times W'}$. For each $b_m = [x_m, y_m, w_m, h_m]$, we map it to feature coordinates (scaling by downsample factor 4) and perform ROI Align [12] to crop a fixed-size patch $v_m \in \mathbb{R}^d$:

$$v_m = ROIAlign(F_1, b_m), \quad (3)$$

Finally, these tokens augment the textual embeddings:

$$F_t^{aug} = \text{concat}(F_t, [v_1; v_2; \dots; v_M]) \in \mathbb{R}^{(K+M) \times d}, \quad (4)$$

where concat denotes tensor concatenation along the row dimension, merging the original text embeddings with the extracted visual tokens to form an expanded matrix. This F_t^{aug} dynamically infuses instance-specific visual semantics into F_t , bridging semantic gaps under domain shifts and enhancing cross-modal alignment for improved query selection and refinement. The entire process is training-free and maintains high efficiency at test time.

Stage 2: Refinement with Augmented Embeddings.

In the second stage of ViTPrompt, we leverage the enriched textual embeddings F_t^{aug} to refine the initial detections through an additional forward pass. Specifically, we replace the original F_t with F_t^{aug} in the query selection and decoding processes, enabling better cross-modal alignment without altering the model’s architecture or parameters.

We begin by recomputing the query indices using the augmented embeddings:

$$\hat{\mathbf{I}} = \text{Top}_N(\text{Max}^{(-1)}(F_v(F_t^{aug})^\top)). \quad (5)$$

This yields a new set of cross-modality queries $\{\hat{q}_j\}_{j=1}^N = F_v[\hat{\mathbf{I}}]$, which are more attuned to the domain-shifted visual features due to the instance-specific visual cues in F_t^{aug} .

Subsequently, the DETR-style decoder refines these queries via interactions with F_v and F_t^{aug} , producing updated query features $\{\hat{q}'_j\}_{j=1}^N$. Each refined feature serves as a proposal embedding $\hat{f}_q \in \mathbb{R}^d$. From these, we compute class logits $\hat{l}_q \in \mathbb{R}^{K+M}$ as:

$$\hat{l}_q = \sigma(\hat{f}_q(F_t^{aug})^\top), \quad (6)$$

Algorithm 1: Visual Token-Augmented Prompting

```
1 Input: Pre-trained Grounding DINO, test images  
    $\{x_i\}_{i=1}^n$ , hyperparameters  $K, \theta, \tau_1, \tau_2$   
2 for each test image  $x_i$  do  
3   Obtain proposals  $\{(b_q, p_q^{init}, s_q)\}_{q=1}^Q$  and  
   largest-scale visual features  $F_1$   
4   Select top- $K$  proposals  
5   Apply class-specific NMS and obtain refined  
   proposals  $\{(b_m, p_m^{init}, s_m)\}_{m=1}^M$   
6   Extract visual tokens  $\{v_m\}_{m=1}^M$  from refined  
   proposals and  $F_1$   
7   Augment textual embeddings:  
    $F_t^{aug} = \text{concat}(F_t, [v_1; v_2; \dots; v_M])$   
8   Recompute query indices:  
    $\hat{\mathbf{I}} = \text{Top}_N(\text{Max}^{(-1)}(F_v(F_t^{aug})^\top))$   
9   Obtain new queries  $\{\hat{q}_j\}_{j=1}^N = F_v[\hat{\mathbf{I}}]$   
10  Decode to get refined proposal embeddings  
    $\{\hat{f}_q\}_{q=1}^N$ , logits  $\hat{l}_q = \sigma(\hat{f}_q(F_t^{aug})^\top)$ , and  
   boxes  $\{\hat{b}_q\}_{q=1}^N$   
11  Calculate  $\bar{l}_q$  and  $\hat{s}_q$  based on  $\hat{l}_q$   
12  Filter high-confidence predictions where  
    $\hat{s}_q \geq \tau_2$  to get  $\{(\hat{b}_q, \hat{p}_q, \hat{s}_q)\}_{q=1}^{\hat{Q}}$  with  
    $\hat{p}_q = \text{softmax}(\bar{l}_q)$   
13 Output: Refined detections
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while bounding boxes $\{\hat{b}_q\}_{q=1}^N \in \mathbb{R}^4$ are regressed from $\hat{f}_q$ using the dedicated head, similar to the initial pass.

To obtain the final predictions, we fuse the expanded logits $\hat{l}_q \in \mathbb{R}^{K+M}$ back to the original K classes. The first K dimensions correspond to the text embeddings for classes 1 to K , while the last M dimensions represent the added visual embeddings, each tied to a class label from its originating prediction $\arg \max(p_m^{init})$. For each class $k \in \{1, \dots, K\}$, we collect the relevant logits—the one from the text embedding $\hat{l}_q[k]$ and those from matching visual embeddings $\{\hat{l}_q[K+m] \mid \arg \max(p_m^{init}) = k\}$ —and select the maximum as the fused logit:

$$\bar{l}_q[k] = \max \left(\{\hat{l}_q[k]\} \cup \{\hat{l}_q[K+m] \mid \arg \max(p_m^{init}) = k\} \right). \quad (7)$$

This yields fused logits $\bar{l}_q[k] \in \mathbb{R}^K$. The score value is simply $\hat{s}_q = \max(\bar{l}_q)$, taken over all $K+M$ dimensions.

High-confidence predictions are filtered where $\hat{s}_q \geq \tau_2$ (potentially adjusted from τ_1), yielding the final set $\{(\hat{b}_q, \hat{p}_q, \hat{s}_q)\}_{q=1}^{\hat{Q}}$, with $\hat{p}_q = \text{softmax}(\bar{l}_q)$. This refinement holistically improves both localization and classification under domain shifts, all in a single extra pass.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate our method on three domain-shift benchmarks for open-vocabulary object detection:

- **PASCAL-C** [26] and **COCO-C** [26] are synthetic corruption benchmarks derived from PASCAL VOC 2007 [7] and MS COCO [21], respectively. Each applies 15 types of corruptions—covering noise, blur, weather, and digital artifacts—at five severity levels to the original test images. Following standard practice [16, 36], we report mean Average Precision at IoU threshold 0.5 (mAP₅₀), averaged over all severity levels per corruption type.
- **FoggyCityscapes** [33] is a realistic adverse-weather dataset generated by synthetically adding fog to the Cityscapes validation set using depth-aware atmospheric scattering models. It contains 500 urban driving scenes annotated with instance-level bounding boxes for 8 common traffic classes (e.g., “car”, “person”, “truck”). Unlike the controlled perturbations in PASCAL-C/COCO-C, FoggyCityscapes captures spatially varying, physics-based domain shifts that closely mimic real-world autonomous driving conditions. We report mAP₅₀ on the full validation set using open-vocabulary prompts without any task-specific tuning.

These benchmarks collectively assess robustness under both synthetic degradations and realistic environmental changes.

Implementation Details. We adopt the official Grounding DINO [22] framework with two visual backbones: Swin-Tiny (Swin-T) and Swin-Base (Swin-B). All models are used in their pretrained, frozen form—no fine-tuning or parameter updates are performed during inference. For hyperparameters, We set the number of visual tokens per object to $K = 5$, the confidence threshold for initial proposal selection to $\theta = 0.6$, and both alignment thresholds in the refinement stage to $\tau_1 = \tau_2 = 0.2$. Following BCA+ [55], the prompt template is set as “[class 1] . [class 2] . . . [class C] .”. All experiments are conducted on a single NVIDIA RTX 2080Ti GPU.

4.2. Results and Analysis

We present a comprehensive evaluation of ViTPrompt on both realistic and synthetic domain-shift benchmarks for open-vocabulary object detection. Our method consistently outperforms state-of-the-art test-time adaptation (TTA) approaches across all settings, without any parameter updates or task-specific tuning.

Performance on FoggyCityscapes. To assess robustness under realistic adverse weather conditions, we first evaluate on FoggyCityscapes [33], a physics-based fog simula-

Table 1. Object Detection Performance Comparison with the state-of-the-art methods on FoggyCityscapes. Metric: Average Precision@50(%); Bp-free: backpropagation-free at test time.

Methods	Venue	Framework	Bp-free	Pson	Rder	Car	Tuck	Bus	Train	Mcle	Bcle	mAP ₅₀
Visual Backbone: ResNet-50												
SHOT [19]	ICML20	Faster RCNN	✗	26.7	30.3	36.9	16.8	28.9	6.4	14.3	23.3	23.0
T3A [13]	NIPS21	Faster RCNN	✗	22.6	23.0	31.9	7.7	14.8	1.0	7.9	19.7	16.6
Self-Training [44]	CVPR21	Faster RCNN	✗	27.7	30.8	41.4	12.8	27.4	4.2	14.8	26.1	23.1
TTAC [36]	NIPS22	Faster RCNN	✗	24.5	27.3	33.4	14.6	26.1	5.8	14.1	21.5	20.9
STFAR[4]	ARXIV23	Faster RCNN	✗	28.8	32.0	42.4	15.1	30.1	11.2	15.5	26.2	25.1
Visual Backbone: Swin-T												
GDINO [22]	ECCV24	Grounding DINO	✓	30.10	3.42	46.26	22.41	31.98	0.08	27.96	28.87	23.88
TDA [16]	CVPR24	Grounding DINO	✓	35.27	4.87	46.43	22.82	32.30	0.38	28.68	29.86	25.08
BCA [56]	CVPR25	Grounding DINO	✓	41.01	5.61	47.85	21.86	33.40	0.91	28.47	29.82	26.12
BCA+ [55]	ARXIV25	Grounding DINO	✓	39.03	3.61	46.98	22.26	35.26	0.87	29.37	35.87	26.65
ViTPrompt	Ours	Grounding DINO	✓	39.25	4.72	50.44	26.17	39.59	0.18	29.39	43.16	29.10
Visual Backbone: Swin-B												
GDINO [22]	ECCV24	Grounding DINO	✓	34.95	20.35	51.56	30.01	45.72	0.87	35.21	32.06	31.34
TDA [16]	CVPR24	Grounding DINO	✓	38.29	25.53	50.09	30.49	45.92	11.98	33.12	39.06	34.31
BCA [56]	CVPR25	Grounding DINO	✓	36.53	24.42	50.77	30.83	45.25	20.29	33.20	42.49	35.47
BCA+ [55]	ARXIV25	Grounding DINO	✓	38.15	26.30	52.30	30.77	45.66	20.29	33.89	42.41	36.22
ViTPrompt	Ours	Grounding DINO	✓	36.54	21.21	53.18	31.96	48.26	23.60	36.78	46.64	37.27

tion benchmark widely used in autonomous driving. As shown in Table 1, ViTPrompt achieves 37.27% mAP₅₀ with the Swin-B backbone, significantly surpassing existing TTA methods: Grounding DINO (31.34%), TDA (34.31%), and BCA+ (36.22%). The gain is especially pronounced for challenging categories such as *Bus* (+2.64) and *Bicycle* (+4.15), where dense fog severely degrades visual appearance. Unlike conventional TTA techniques that only adjust confidence scores or feature statistics, ViTPrompt *generates new classification hypotheses* by injecting RoI-aligned visual tokens into the language prompt, enabling the detector to reinterpret ambiguous regions based on localized visual evidence.

Performance on PASCAL-C. We evaluate ViTPrompt on the PASCAL-C corruption benchmark [26]. As shown in Table 2, our method achieves state-of-the-art performance across all corruption types. With the Swin-T backbone, ViTPrompt attains an average mAP₅₀ of 43.03%, surpassing BCA+ by 2.36 points, with consistent gains across all 15 corruptions (e.g., +3.05 on *frost*). Using the stronger Swin-B backbone, it further advances the average to 71.40% (+2.09 over BCA+), delivering significant improvements under challenging distortions like *glass blur* (+3.31) and *Gaussian noise* (+1.75). These results confirm the effectiveness of our visual token-driven refinement in enhancing detection robustness under diverse corruptions.

Performance on COCO-C. We further evaluate ViTPrompt on COCO-C [26], a challenging benchmark comprising 15 diverse corruptions applied to the COCO validation set. As shown in Table 3, our method consistently

outperforms all prior approaches across all corruption types and backbones. With the Swin-T backbone, ViTPrompt achieves an average mAP₅₀ of 26.25%, surpassing the previous best method BCA+ (25.06%) by 1.19 points. Although the absolute margin may appear modest, it is noteworthy given the difficulty of adapting to 80 categories under severe and varied distortions—more importantly, ViTPrompt establishes a new state of the art in *every single corruption type*. For instance, it improves performance under *snow* (+1.36), *Gaussian noise* (+1.17), and *motion blur* (+0.64). When equipped with the stronger Swin-B backbone, ViTPrompt further advances the results to 42.28% mAP₅₀, exceeding BCA+ (39.98%) by 2.30 points. The gains are again universal across corruptions; notable improvements include +2.40 under *snow* (47.85 → 50.25), +1.97 under *motion blur* (31.72 → 33.69), and +3.23 under *Gaussian noise* (34.04 → 37.27). These consistent enhancements stem from our core mechanism: by injecting RoI-aligned visual tokens into the prompting process, ViTPrompt dynamically refines semantic representations at test time, enabling robust label assignment even when low-level features are severely corrupted.

Analysis of Backbone Scaling. Both ViTPrompt and baselines benefit from stronger backbones (Swin-T → Swin-B), but ViTPrompt exhibits superior scalability. On PASCAL-C, the performance gain from Swin-T to Swin-B is +28.37 for ViTPrompt versus +28.64 for BCA+—nearly identical in absolute terms, but ViTPrompt starts from a higher baseline and maintains a consistent lead. More importantly, on COCO-C, ViTPrompt’s improvement (+16.03) exceeds that of BCA+ (+14.92), suggesting better utilization of

Table 2. Object Detection Performance Comparison with the state-of-the-art methods on PASCAL-C. Metric: mean Average Precision@50(%).

Methods	Brit	Contr	Defoc	Elast	Fog	Frost	Gauss	Glass	Impul	Jpeg	Motn	Pixel	Shot	Snow	Zoom	Average
Visual Backbone: ResNet-50																
SHOT [19]	72.0	31.7	18.9	46.6	67.5	45.8	12.0	11.6	16.4	41.8	19.7	33.1	19.9	42.5	27.6	33.8
T3A [13]	36.9	12.5	11.0	19.7	32.7	20.6	6.1	6.4	6.5	14.8	10.1	13.2	8.4	16.8	13.8	15.3
Self-Training [44]	67.9	39.3	2.6	52.5	65.7	47.2	11.9	20.2	12.1	29.3	4.1	6.9	17.4	44.9	9.5	28.8
TTAC [36]	72.2	40.4	29.3	58.1	68.7	50.4	29.8	28.7	33.6	46.4	29.2	46.1	35.1	48.0	34.9	43.4
STFAR [4]	67.3	51.8	34.8	55.7	65.2	50.7	32.4	34.6	36.3	49.4	34.6	55.7	37.8	50.9	34.8	46.1
Visual Backbone: Swin-T																
GDINO [22]	63.04	45.66	35.58	31.33	60.92	50.52	25.82	18.84	27.99	39.34	26.99	13.39	30.49	45.89	23.38	35.95
TDA [16]	64.61	47.71	37.33	32.41	61.77	52.19	27.05	20.48	29.35	41.45	28.50	14.15	31.85	48.07	24.35	37.42
BCA [56]	68.11	49.42	38.46	34.42	64.27	53.21	28.37	22.24	31.90	42.81	30.39	14.81	33.68	49.37	24.97	39.10
BCA+ [55]	70.25	50.72	40.79	37.08	67.83	56.38	28.59	22.48	31.74	47.40	30.75	14.87	33.66	51.87	25.65	40.67
ViTPrompt	72.35	52.37	43.16	40.07	70.49	59.43	32.65	24.77	34.68	48.04	33.38	16.70	36.78	53.95	26.71	43.03
Visual Backbone: Swin-B																
GDINO [22]	85.16	70.69	56.12	55.56	84.56	73.75	54.94	41.17	57.72	71.44	56.08	66.17	60.54	76.48	36.94	63.15
TDA [16]	86.28	74.33	57.98	58.45	85.98	76.41	57.82	43.84	60.80	74.31	57.91	69.45	63.79	78.36	38.84	65.64
BCA [56]	88.66	75.73	59.84	61.00	88.02	77.69	60.21	45.98	62.46	76.32	60.55	71.22	64.89	80.67	40.66	67.59
BCA+ [55]	89.39	77.85	62.04	62.71	88.87	79.94	61.81	47.67	64.87	78.12	61.81	73.51	67.28	82.42	41.42	69.31
ViTPrompt	90.49	80.12	65.19	65.04	90.14	81.17	63.56	50.98	67.37	80.02	63.53	75.20	71.31	83.87	43.11	71.40

Table 3. Object Detection Performance Comparison with the state-of-the-art methods on COCO-C. Metric: mean Average Precision@50(%).

Methods	Brit	Contr	Defoc	Elast	Fog	Frost	Gauss	Glass	Impul	Jpeg	Motn	Pixel	Shot	Snow	Zoom	Average
Visual Backbone: ResNet-50																
SHOT [19]	40.9	26.6	14.7	19.7	41.5	26.7	11.0	7.2	12.1	16.4	11.0	9.7	13.0	22.0	6.4	18.6
T3A [13]	28.8	15.9	8.3	11.3	28.9	17.2	4.6	3.1	5.2	9.0	5.8	4.1	5.8	13.8	3.5	11.0
Self-Training [44]	38.1	28.4	14.7	25.5	38.5	27.9	16.7	11.4	18.8	23.8	16.0	24.5	18.6	27.6	7.8	22.6
TTAC [36]	38.3	29.5	15.1	28.2	39.0	28.5	16.8	14.3	18.0	23.2	14.3	24.8	19.3	26.7	8.7	23.0
STFAR [4]	39.1	31.1	16.8	29.0	39.0	29.2	19.2	15.4	20.1	26.1	17.2	28.3	21.0	29.5	10.2	24.7
W3TTAOD [47]	36.4	27.2	14.0	27.2	37.4	27.2	13.6	13.6	16.1	22.3	14.2	22.2	16.6	23.7	8.3	21.3
Visual Backbone: Swin-T																
GDINO [22]	42.91	23.83	19.45	23.76	43.26	30.36	15.03	10.22	15.87	23.77	16.56	7.31	16.99	25.51	9.35	21.61
TDA [16]	45.53	24.43	20.33	27.08	45.91	32.31	15.77	9.93	16.72	26.25	16.85	8.99	17.10	28.22	9.62	23.00
BCA [56]	46.34	25.18	21.03	27.45	46.74	33.92	15.97	10.91	17.33	26.19	16.05	8.89	17.97	29.02	9.87	23.52
BCA+ [55]	49.62	26.56	22.20	28.65	49.79	35.65	17.97	11.47	18.67	28.38	19.10	7.83	19.86	30.10	10.08	25.06
ViTPrompt	52.01	27.22	23.06	29.95	51.93	37.10	18.94	12.43	20.35	28.38	19.74	10.16	21.17	31.46	9.94	26.25
Visual Backbone: Swin-B																
GDINO [22]	55.36	40.40	29.11	36.51	56.26	44.39	30.27	21.38	31.30	40.26	28.78	36.68	32.67	42.71	13.44	35.97
TDA [16]	56.86	42.26	30.35	38.15	58.21	46.72	32.28	22.64	32.84	42.12	29.94	38.97	34.17	44.14	14.89	37.64
BCA [56]	57.21	42.32	31.00	38.90	58.84	46.43	32.17	23.37	33.53	42.49	31.15	39.56	35.76	45.42	14.92	38.20
BCA+ [55]	60.34	44.99	32.25	40.85	60.92	49.06	34.04	24.42	35.33	44.91	31.72	41.17	36.59	47.85	15.24	39.98
ViTPrompt	63.19	46.75	33.66	42.57	63.96	51.73	37.27	26.58	38.44	46.32	33.69	44.39	39.74	50.25	15.67	42.28

high-capacity features for complex scenes. This trend confirms that ViTPrompt’s region-aware prompting mechanism scales effectively with model capacity.

In summary, ViTPrompt achieves state-of-the-art results on both realistic (FoggyCityscapes) and synthetic (PASCAL-C/COCO-C) domain-shift benchmarks. Crucially, its advantage stems not from post-hoc calibration, but from *test-time generation of refined classification proposals* via visual-language fusion—a capability absent in prior TTA methods.

4.3. Robustness and Ablation Studies

We conduct four ablation studies to analyze the design choices and robustness of ViTPrompt.

Ablation on Visual Prompt Fusion (Eq. 7). The core of

Table 4. Ablation on the prompt fusion strategy in Eq. 7. “Text-only” uses the original language embedding without visual feedback; “ViTPrompt (max)” applies max-pooling over RoI-aligned visual tokens to refine the prompt. Results use Swin-T backbone.

Method	FoggyCityscapes	COCO-C (avg)
Text-only (no visual fusion)	28.89	25.57
ViTPrompt (max fusion, Eq. 7)	29.10	26.25

ViTPrompt lies in Eq. 7, where RoI-aligned visual tokens are fused with the original text embedding via max-pooling to construct an instance-adaptive prompt. To validate this design, we compare it against a baseline that forgoes visual feedback and relies solely on the static language embedding—equivalent to standard Grounding DINO at test time.

Table 5. Main results on domain-shifted benchmarks. We report COCO-style mAP@0.5:0.95 using a Swin-T backbone.

Method	FoggyCityscapes	COCO-C (avg)
Grounding DINO	15.65	15.27
ViTPrompt (Ours)	19.02	18.36

As shown in Table 4, integrating visual context through max-pooling consistently improves performance, yielding gains of +0.21 mAP on FoggyCityscapes and +0.68 mAP on COCO-C (average across corruptions). These results demonstrate that dynamically refining textual prompts with visual evidence enhances robustness under domain shift, confirming the value of adaptive prompting over fixed language queries in open-vocabulary detection.

Improved Localization via Visual Prompt Refinement.

Unlike conventional open-vocabulary detection methods that score proposals using fixed text embeddings, ViTPrompt leverages instance-specific visual cues to refine both classification scores and, crucially, *bounding box quality*. As shown in Table 5, our method achieves significant gains in mAP@0.5:0.95—+3.37 points on FoggyCityscapes and +3.09 points on COCO-C—highlighting its superior robustness under domain shift. The improvement stems from our two-stage design: by fusing RoI-aligned visual tokens with the original text prompt (Sec. 3.2), ViTPrompt generates dynamic, instance-adaptive prompts that better capture the true spatial extent of objects. This enables the model to re-score and effectively correct coarse or misaligned boxes produced by the initial Grounding DINO pass. In contrast to prior works that only modulate semantic representations, our approach explicitly enhances *localization accuracy* through visual grounding, which is key to the observed performance gains.

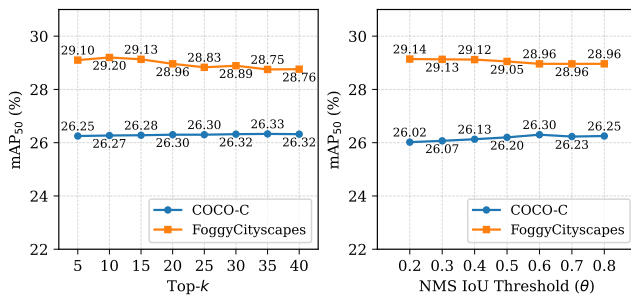


Figure 3. Parameter sensitivity analysis on COCO-C (average mAP₅₀) and FoggyCityscapes using Swin-T backbone. (a) Performance vs. number of top- K RoI-aligned visual tokens used in prompt fusion. (b) Performance vs. NMS IoU threshold θ . ViTPrompt is robust across a wide range of k and θ , with optimal settings at $K = 10$ and $\theta = 0.6$.

Sensitivity to Top- K Visual Tokens. We study how the number of top- K RoI-aligned visual tokens selected for

prompt fusion affects performance. As shown in Figure 3(a), ViTPrompt achieves peak mAP when $K = 10$, with graceful degradation for smaller or larger K . This indicates that a moderate set of salient visual cues is sufficient for effective language refinement, while too few tokens lack discriminability and too many introduce noise.

Sensitivity to NMS Threshold θ . We also analyze the impact of the non-maximum suppression (NMS) IoU threshold θ during inference. Figure 3(b) shows that ViTPrompt performs best at $\theta = 0.6$, balancing duplicate removal and recall. Notably, our method remains stable across $\theta \in [0.2, 0.8]$, confirming that performance gains are not due to aggressive post-processing.

Table 6. Efficiency comparison (Swin-T backbone, input size 600×600 , batch size=1).

Method	#Params (M)	FLOPs (G)	GPU Memory (MB)
Grounding DINO [22]	58.30	145.64	2946
ViTPrompt (Ours)	58.35	252.26	2946

Computational and Memory Overhead. We report efficiency metrics for Grounding DINO and ViTPrompt with a Swin-T backbone under identical settings (input size 600×600 , batch size=1) in Table 6. While ViTPrompt introduces only a marginal increase in model parameters (+0.05M), its FLOPs rise to 252.26G due to RoI-aligned visual token extraction and prompt refinement. Notably, both methods exhibit identical peak GPU memory consumption (2946 MB), indicating that our instance-level prompting operates within the same memory budget as the base detector—thanks to on-the-fly RoI feature reuse without additional caching. This confirms that ViTPrompt achieves substantial robustness gains with negligible deployment overhead.

5. Conclusion

We present ViTPrompt, a training-free test-time adaptation framework for open-vocabulary object detection. By dynamically refining text prompts with instance-level visual tokens, ViTPrompt aligns representations with true object extents without model updates or training data. This plug-and-play strategy significantly enhances localization accuracy and robustness under domain shifts, offering a simple yet effective solution for real-world deployment.

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